Project Brief

Thermal Sensitivity of Ant Foraging in a Northern Sicilian Olive Grove: Behavioural Responses and Implications for Olive Production

Applicant

Dean Hart, BSc (Hons) Natural Sciences (Biology); PhD in Research (Clinical Psychology)
Termini Imerese (PA), Sicily, Italy
Resident landholder managing an organically run olive estate suitable for ecological fieldwork.

1. Background

Mediterranean agroecosystems are experiencing increasingly frequent and intense heat events, with Northern Sicily representing a climate-change hotspot where olive cultivation remains economically and culturally significant. Ants are key components of these systems, contributing to pest suppression, seed dispersal, soil turnover, and nutrient cycling. Their foraging behaviour is strongly influenced by temperature, shaped by neuromuscular constraints, metabolic limits, and behavioural thermoregulation.

Despite their ecological importance, little is known about how rising temperatures affect the activity patterns of ant species commonly found in olive landscapes. As heat events intensify, shifts in ant foraging behaviour may alter ecosystem functions that indirectly support olive production. Understanding these behavioural responses is essential for predicting how warming may reshape ecological processes in Mediterranean agricultural systems.

2. Research Aim

To investigate how elevated temperatures influence the foraging behaviour of ant species in a Northern Sicilian olive grove, and to assess the potential implications of these behavioural shifts for olive production.

3. Research Questions

1. How do foraging activity levels of different ant species vary across the full daily temperature range in a Northern Sicilian olive grove?
2. Do species differ in their thermal thresholds for initiating, maintaining, or ceasing foraging?
3. How does micro-position on the tree (ground, trunk, canopy) mediate species-specific responses to rising temperatures?
4. What are the potential implications of these behavioural shifts for ecosystem functions relevant to olive production?

4. Proposed Methodology

Fieldwork will be conducted in an organically managed olive grove in Termini Imerese, Northern Sicily. The site provides a stable, accessible environment for repeated behavioural observations.

• Sampling design — Select focal trees and active foraging trails representing the dominant ant species present.
• Behavioural observations — Conduct repeated focal sampling across the full daily temperature cycle, including early morning, midday heat peaks, and evening decline. Record activity levels, foraging intensity, and behavioural states.
• Micro-position analysis — Quantify behaviour at three vertical zones: ground, trunk, and canopy, allowing assessment of behavioural thermoregulation and microhabitat buffering.
• Temperature measurements — Use fine-scale temperature loggers to record ambient and micro-position temperatures concurrently with behavioural observations.
• Data analysis — Compare species-specific thermal sensitivity, identify behavioural thresholds, and model activity patterns relative to temperature gradients.

This methodology is low-cost, feasible, and well-suited to a single-researcher field project with strong potential for publication.

5. Expected Outcomes

• Comparative datasets on thermal sensitivity across ant species in a Mediterranean olive system.
• Identification of behavioural thresholds associated with heat exposure.

• Evidence of species-specific shifts in foraging windows under elevated temperatures.
• Insights into how warming may alter ant-mediated ecosystem functions such as pest suppression and nutrient cycling.

These findings will contribute to climate-change biology, behavioural ecology, and agroecology, with practical relevance for olive-grove management.

6. Significance

This project addresses urgent questions about how climate change reshapes ecological interactions in Mediterranean agricultural landscapes. Ants play a critical role in olive agroecosystems, yet their behavioural responses to rising temperatures remain poorly understood.

By generating regionally relevant data from a Northern Sicilian olive grove, this research will:

• improve understanding of climate-driven behavioural adaptation
• inform sustainable olive-grove management
• support climate-resilient agricultural strategies
• contribute to broader ecological models of thermal stress in invertebrates

The project aligns with current priorities in environmental change research and offers strong potential for publication and applied impact.

7. Applicant Suitability

Dean Hart holds a PhD in Research (Clinical Psychology), awarded for work examining trauma, emotional regulation, and behavioural adaptation. This complements his biological research by deepening his understanding of stress processes and behavioural responses. He also holds a BSc (Hons) in Natural Sciences (Biology) with distinctions in neurobiology and applied biological research.

As a resident landholder with direct access to the study site, he is well positioned to conduct consistent fieldwork and long-term monitoring. His academic training in biology, behavioural ecology, and environmental stress provides a strong foundation for the proposed research, and his independent research experience ensures the project is feasible, well-structured, and likely to produce publishable outcomes.